PeriGen
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PeriWatch Vigilance™ Tracings
Interface Specification

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1 Vigilance Interface Specification

The purpose of this section is to outline the PeriWatch Vigilance Tracings (**PWV**) expectations for HL7 message exchange with other systems. This document should form the foundation for interface design discussions and presumes prior knowledge of the Health Level Seven (**HL7**) messaging standard; details available at http://www.hl7.org.

1.1 Communications

The PWV integrations engine resides on PWV training and production application servers, there sharing server IPs, network domain authorization/restrictions, and server hardware and database with the related application. As hardware and database are shared between application and integrations engine, PWV expresses strong preference towards the configuration of a **client-side filter**, where possible only passing messages related to existing or potential PWV patients to our engine for consideration and consumption. Often this filtering is accomplished by logic parsing PV1.3.1, passing messages only for patients with an assigned location of (e.g.) LND, LDRP, and/or OP OB Clinic; specific implementation of these filters is at client discretion. Additional requisite filtering may be required; details below where relevant.

Interface messaging occurs across TCP/IP connections. Our standard implementation port for incoming messages is **17676**. Outbound message destinations are expected to be the IP and port of a receiving interface engine; however, data may also be placed on a local file structure in various formats for collection at client preference.

PWV is configured for serialized/bimodal message handling, transmitting or waiting for reception of acknowledgment messages immediately after message reception. Failed transmissions will retry or wait for retry until five attempts have passed, then closing the port and marking the route in error.

The low-level format of the HL7 data stream adheres to standard:

<SB>dddd**<EB><CR>**

…where **<SB>** is the start block character (1 byte ASCII <VT>, 0x0B),
dddd represents the HL7 message data stream,
<EB> is the end block character (1 byte ASCII <FS>, 0x1C), and
<CR> is the carriage return character (0x0D).

1.2 Interface Description

HL7 Message Definitions

The PWV interface standard adheres to the published HL7 standards for any v2.x message type. Valid transmitted message segments which are not mentioned in the following message definitions are ignored by our application.

Message delimiters adhere to standard: |, ^, &, ~, and \ being reserved characters to delimit fields, components, subcomponents, repeating fields, and escaped characters (respectively).

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PWV currently has the capacity to process inbound HL7 messages pertaining to registration (ADT), unsolicited results (ORU). Inbound data from host EHR systems are accepted as specially-formatted ORU unsolicited result (R01) messages and/or inside OBX message segments appended to ADT messages.

The following sections define the required and optional components of PWV HL7 messages. Indentations denote logical message sub-blocks; fields in curly braces (**{}**) may repeat, those in square brackets (**[]**) are optional per specification.

1.3 Admit, Discharge, Transfer (ADT)

PWV only processes received Admit, Transfers, Discharges, and Update (A01, A02, A03, and A08, respectively) messages; all others are ignored and should be filtered from transmission.

Message Structures

ADT^A0{1,2,3,8}	Admit, Transfer, Discharge, Update
MSH	Message Header
EVN	Event Type
PID	Patient Identifier
NK1	Next-of-Kin / Emergency Contact
PV1	Patient Visit Information
[{ **PV2** }]	Additional Patient Visit Information
[{ **OBX** }]	Observation Segments
[{ **AL1** }]	Patient Allergy Information
[{ **DG1** }]	Patient Diagnoses

ADT Example

The following section describes the expected behavior of a “standard” sample of ADT HL7 message.

The following message was run through the interface engine into PWV (message formatted for legibility):

```
MSH|^~\\&|SendApp|SendFac|RecvApp|RecvFac|201101010001||ADT^A01|ADT201947277|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN STREET^^TRUSTYVILLE^DC^20202||(202)202-
0202^^^^^202^2020202||SIGN|SINGLE|NO PREFERENCE OF RELIGION|987654321|111-22-3333||||||||||
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)2020203^^^^^202^2020204
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
SERVICE||||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|
```

1.4 Vigilance Inbound Discrete Data

Inbound assessment data is expected per the described ORU^R01 message container.

ORU^R01	Unsolicited Results
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
{	
OBR	Observation Request
{	
OBX	Observation / Result
[{ **NTE** }]	Notes and Comments
[**SPM**]	Specimen
}	
}	

Assessments themselves are to be passed in OBX segments (which may repeat). An assessments unique identifiers are the patient's medical record number, the datetime of the assessments (with resolution fixed at "second", 120100.000 and 120100.999 would be interpreted as occurring at the same time), and the field prefix.

OBX|n||ANNOTATION EXAMPLE||SOME TEXT HERE.|||||F|||20121212121202||USER

DateTime of the assessment must be in local time of the PWV server, as we perform an internal conversion to UTC based on the PWV application's server time.

Available Vigilance inbound assessment data include:

<u>Fields that feed Vigilance Curve:</u>

• EDD/LMP • Gestational Age • Gravida • Para • Term • Pre-Term • AB Induced • AB Spontaneous • Living	• Previous Vaginal Delivery • Attempted VBAC • Number of Fetuses • Dilation • Effacement • Station • Presentation • Epidural in Use (DateTime)

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Vital signs from the EMR:

- Maternal Heart Rate
- Maternal BP
- Maternal SpO2
- Respiratory Rate
- Temperature

1.5 Vigilance Outbound Notification(s)

Outbound notification(s) are transmitted utilizing the previously described ORU^R01 container. Vigilance will transmit this message to the EMR when the Vigilance status changes. This value will typically be stored in a new flowsheet record created within the EMR.

NOTE: The Vigilance states sent via HL7 are intended to be used by the clinical team to display and color code these values on the EMR relevant screens

For outbound notifications, OBX.3 will contain the notification(s) status of the patient; OBX.5 will hold the value – e.g.,

OBX|2|ST|State||Notified||||||

Available Notification status include:

- State
- Reason
- URI (for future use, currently not in the outbound message)

1.5..1 Notified

MSH|^~\&|VIGILANCE|LND|LND|InterfaceEngine|20181019134800||ORU^R01|ORU20181019134808|D|2.5.1
PID|||1433||onepatient^onepatient||19821019||||||||||1
OBR|||999999|NotificationStatus^Flowsheet|||00000000000000
OBX|1|TX|STATE||Notified||||||F
OBX|2|TX|REASON||Cues State fetus:2. GA not documented. Cues only validated for 36w or more, Curve for 35w or more.||||||F

1.5..2 Not Notified

MSH|^~\&|VIGILANCE|LND|LND|InterfaceEngine|20181019134700||ORU^R01|ORU20181019134758|D|2.5.1

PID|||23422||twopatient^twopatient||19821019||||||||||1
OBR|||999999|NotificationStatus^Flowsheet|||00000000000000
OBX|1|TX|STATE||NotNotified||||||F
OBX|2|TX|REASON||||||||F

1.5..3 Error

MSH|^~\&|VIGILANCE|LND|LND|InterfaceEngine|20181019134800||ORU^R01|ORU20181019134803|D|2.5.1
PID|||45444||threepatient^threepatient||19821019||||||||||1
OBR|||999999|NotificationStatus^Flowsheet|||00000000000000
OBX|1|TX|STATE||Error||||||F
OBX|2|TX|REASON||Notification Service Connection Failure||||||F

Segment and Field Definitions

The following tables describe fields directly consumed by PWV. The intention of this section is to describe which fields are directly mapped into (or out of) the PWV application. Fields required by the HL7 standard specification are still required for a message to be considered valid: omission of a field otherwise required by the HL7 standard specification from tables below indicates only that it is otherwise ignored by the PWV interface engine logic.

Unless otherwise noted, any additional subfield breakdowns will default to the first segment (e.g., if XXX.3.2 is noted in the table below but a given message defines subfields within XXX.3.2, XXX.3.2.1 will be utilized by default). If a field may repeat per specification, this field will be notated within the table by either a subscript **0** or **n;** indicating our use of only the first of the repeats and ignoring all subsequent fields or processing all passed repeats, respectively. Thus $AL1_n.5_0$ indicates that all passed allergy segments are processed, but that only the first .5 segment is considered for each AL1 – any subsequent .5 segments in a given AL1 message would be ignored.

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AL1$_n$			**Allergy Information**
.2	Type	**.3.2**	Code, Mnemonic, Description (Text)
.3.1	Code, Mnemonic, Description (Identifier)	**.5$_0$**	Reaction
DG1$_0$			**Diagnosis**
.3	Code	**.4**	Description
MSH			**Message Header**
.3	Sending Application	**.9**	Message Type and Trigger
.4	Sending Facility	**.10**	Control ID
.5	Receiving Application	**.11**	Processing ID
.6	Receiving Facility	**.12**	Version ID
.7	Date/Time of Message		
NK1$_0$			**Next-of-Kin**
.2	Contact Name	**.5$_0$.1**	Phone Number
.3	Relationship		
NTE			**Notes and Comments**
.1	Set ID	**.3$_n$**	Comment
OBR			**Observation Request**
.3	Filler Order Number	**.4.2**	Universal Service ID (Text)
.4.1	Universal Service ID (Identifier)	**.7**	Observation Date/Time
OBX$_n$			**Observation / Result**
.3.1	Observation Identifier (Identifier)	**.11**	Observation Result Status { P, F, C }
.3.2	Observation Identifier (Text)	**.23**	Performing Organization's Name
.3.4	Observation Identifier (Alternate ID)	**.24.1**	Performing Organization's Street Address
.5$_n$	Observation Value	**.24.3**	Performing Organization's City
.6.2	Units (Text)	**.24.4**	Performing Organization's State
.7	References Range	**.24.5**	Performing Organization's Zip Code
.8$_0$	Abnormal Flags		
ORC			**Common Order**
.1	Order Control	**.7.2.2**	Quantity Timing, Explicit Time Interval
.3	Filler Order Number	**.9**	Date/Time of Transaction

PID			**Patient Identification**
.3$_0$.1	Internal Patient ID (Medical Record Number)	**.11$_0$.4**	Patient's Address – State
.5$_0$.1	Family Name	**.11$_0$.5**	Patient's Address – Zip Code
.5$_0$.2	Given Name	**.13$_0$.1**	Home Phone Number
.5$_0$.3	Middle Name or Initial	**.15.1**	Primary Language (Identifier)
.7	Date/Time of Birth	**.16.1**	Marital Status (Identifier)
.10$_0$.1	Race (Identifier)	**.17.1**	Religion (Identifier)
.11$_0$.1	Patient's Address – Street Address	**.18.1**	Patient Account / Visit Number
.11$_0$.3	Patient's Address – City	**.21$_0$.1**	Mother's Medical Record Number
PV1			**Patient Visit**
.3.3	Assigned Patient Location (Bed)	**.17.3**	Admitting Doctor's Given Name
.9.2	Consulting Doctor's Family Name	**.44**	Admission Date/Time

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.9.3	Consulting Doctor's Given Name	**.52.2**	Attending Nurse's Family Name
.17.2	Admitting Doctor's Family Name	**.52.3**	Attending Nurse's Given Name
PV2			**Patient Visit – Additional Information**
.3.2	Reason for Admission		

2 Technical Support

PeriGen® IT Solutions Technical Support is available 24/7, 365 days a year. If you need help, please call: 1-855-236-4353. support@perigen.com

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